

Bangzhao Shu

Medford, MA 02155 | +1 734-210-3367 | shu.b@northeastern.edu | [Google Scholar](#) | [GitHub](#) | [Homepage](#)

Third-year Computer Science Ph.D. student at Northeastern University specializing in NLP. My research lies at the intersection of affective NLP, mechanistic interpretability, and computational social science, with a focus on evaluating and improving the social and emotional intelligence of large language models.

EDUCATION

PhD in Computer Science, Northeastern University	2024-Present
• <i>Research Areas: NLP / Large Language Models, AI Safety, Mechanistic Interpretability, Computational Social Science</i>	
Master of Science in Information, University of Michigan	2022-2024
Master of Science in Environment and Sustainability, University of Michigan	2021-2023
Bachelor of Engineering in Geomatics, Nanjing Normal University	2016-2020

PUBLICATIONS

Peer-Reviewed Papers

- [1] **EMNLP 2026 (accepted)** [Paper](#) (first author): Used sparse autoencoders to identify and causally validate internal features explaining why LLMs recognize some emotions better than others; uncovered emotion-specific representational structures and improved overall recognition through targeted feature interventions.
- [2] **ICWSM 2026** [Paper](#) (first author): Constructed EXPRESS, a fine-grained emotion recognition benchmark with 250+ emotion categories grounded in psychological theories. Evaluated 14 language models using various prompting strategies, and found that LLMs perform poorly on fine-grained emotion recognition.
- [3] **ACM Transactions on Social Computing (accepted)** [Paper](#): Studied how mothers experiencing burnout use Reddit to share their experiences, analyzed the support they receive from the community, and examined co-parenting dynamics as a contributing factor.
- [4] **NAACL 2025** [Paper](#) (co-first): Released SIZZLER, a dataset of 1,800 annotated emails capturing nuanced intents and expectations, benchmarked models from logistic regression to zero-shot LLMs, and conducted large-scale causal analysis on 11.3M emails to examine how linguistic and social factors shape response dynamics.
- [5] **NAACL 2024** [Paper](#) (co-first): Constructed MODEL-PERSONAS, a benchmark comprising 39 psychological instruments for evaluating LLM personas, and assessed the personas and their stability across 17 LLMs, revealing notable persona instability.

Papers Under Review

- [6] **Submitted to Nature Mental Health** (first author): Introduced HEAT, a framework for assessing systematic biases in LLM emotion recognition, and revealed consistent biases toward negative and more intense emotions across topics and models.

RESEARCH EXPERIENCE

CSG Lab , Northeastern University, Advisor: Mai ElSherief	2024-Present
<i>Affective NLP, Computational Social Science, and Mechanistic Interpretability</i>	
• Study how LLMs represent and recognize emotion across cultural contexts through benchmark development, bias analysis, and mechanistic interpretability.	
Blablalab , University of Michigan, Ann Arbor, Advisor: David Jurgens	2023-2025
<i>Computational Social Science, Human-Centered NLP, and LLM Evaluation</i>	
• Developed large-scale NLP datasets and models for studying LLM personas, online communication, and speech prosody.	
GISense Lab , The University of Texas at Austin, Advisor: Yuhao Kang	2023-2024
<i>GeoAI, Supervised Fine-Tuning</i>	
• Fine-tuned LLMs to generate and enrich semantic information from geospatial data.	

WORK EXPERIENCE

<i>Graduate Student Intern</i> , Cooperative Institute for Great Lakes Research	2021-2023
• Developed generalized linear models to identify key factors influencing chlorophyll and microcystin concentrations, and evaluated the impact of sediment resuspension on predictive performance.	

SKILLS

- **Programming Languages:** Python, R, SQL
- **ML & NLP Frameworks:** PyTorch, Hugging Face, Scikit-learn, PEFT, SentenceTransformers, LangChain
- **Data & Analysis:** Large-Scale Data Processing, Statistical Analysis
- **Research Methods:** LLM Evaluation & Benchmarking, Mechanistic Interpretability, Causal Inference, Supervised Fine-Tuning